

Marketing A/B Testing

Campaign Performance Dashboard

2022 | MySQL + Power BI

588,101 Users · 14,843 Conversions · 2.52% Conv Rate · 43.09% Ad Lift · 2 Groups

TOOLS	SOURCE	TEST GROUP's	EXCEL USED	RECORDS
MySQL + Power BI	Kaggle (Public)	ad (96%) · psa (4%)	No — Not Required	588,101 · 6 Cols

01 EXECUTIVE SUMMARY

- Conducted thorough EDA on 588,101 marketing A/B testing records — confirming zero nulls, zero duplicates, zero duplicate user IDs, and determining Excel was NOT required — loaded CSV directly into MySQL via a single `LOAD DATA LOCAL INFILE` command that simultaneously skipped the row-index column (Unnamed: 0) and converted boolean True/False to integer 1/0 for the converted column, completing in 6.922 seconds.
- Executed 7 analytical SQL queries covering overall KPIs, ad vs PSA group comparison, conversion by day, conversion by hour, ads exposure bucket analysis, best day+hour combinations, and created `vw_ab_summary` VIEW — all without any import errors, data quality issues, or re-imports.
- Connected Power BI live to MySQL, created 8 DAX measures, and built a 2-page dashboard (16 visuals) with Sync Slicers — Page 1 (Campaign Overview) showing group performance and KPIs, Page 2 (Timing & Exposure Analysis) showing day, hour, heatmap, and ads bucket analysis.
- Confirmed ads are 43.09% more effective than PSA — ad group conversion rate 2.5547% vs PSA 1.7854% across 564,577 vs 23,524 users respectively; identified Monday 14:00–17:00 as optimal targeting window (4.44%) and a direct relationship between ad exposure and conversion — users seeing 101-200 ads convert at 17.37% vs 0.33% for users seeing 1-10 ads.

02 PROBLEM STATEMENT

A marketing team ran an A/B test exposing 588,101 users to either real advertisements (ad group) or public service announcements (PSA group — control) to determine whether their ad campaign generated statistically meaningful conversion lift over the baseline. With data stored as a raw CSV file and no reporting layer, the team could not quantify the campaign's effectiveness, identify optimal ad delivery timing, understand how ad exposure frequency affected conversion probability, or determine which day-hour combinations maximised return on ad spend — leaving a significant campaign optimisation opportunity unmeasured across 588,101 user interactions.

03 OBJECTIVES

- Obj 1: Perform complete EDA to determine exact data quality status and assess whether Excel pre-processing is necessary before MySQL import.
- Obj 2: Import 588,101 records directly into MySQL via `LOAD DATA LOCAL INFILE` — skipping the row-index column and converting boolean converted values to integer 0/1 in a single command.

- Obj 3: Execute 7 analytical SQL queries quantifying A/B test results, timing patterns, exposure effects, and optimal targeting windows.
- Obj 4: Create vw_ab_summary VIEW pre-aggregating data by test group, day, hour, and ads bucket for Power BI performance.
- Obj 5: Build a 2-page Power BI dashboard with 8 DAX measures and 16 visuals covering campaign overview, group comparison, timing heatmap, and exposure analysis.
- Obj 6: Deliver actionable recommendations — best timing window, optimal exposure frequency, and campaign lift quantification for future ad budget allocation.

04 DATASET OVERVIEW & EDA FINDINGS

Attribute	Detail
Dataset Name	Marketing A/B Testing Dataset
Source	Kaggle — faviovaz/marketing-ab-testing
File Format	CSV (.csv)
Raw Records	588,101 rows · 7 columns (Unnamed:0 index col dropped at import)
Clean Records	588,101 — zero nulls, zero duplicates, zero duplicate user IDs
Date Coverage	No date column — data represents a single campaign period
Test Groups	ad: 564,577 users (96.00%) · psa: 23,524 users (4.00%)
Null Values	Zero across all 6 columns ■
Duplicates	Zero exact duplicates · Zero duplicate user IDs ■
Excel Required	NO — data loaded directly CSV → MySQL
Key EDA Finding	total_ads right-skewed (mean 24.82, median 13, max 2,065); 8.85% IQR outliers — all kept as valid real users

Column Definitions (6 Columns — after dropping Unnamed:0)

Column	MySQL Type	Description
user_id	INT PRIMARY KEY	Unique user identifier — 588,101 distinct values, no duplicates
test_group	VARCHAR(3)	Test assignment: 'ad' (real ad, 564,577 users) or 'psa' (control, 23,524 users)
converted	TINYINT(1)	Conversion outcome: 1=converted, 0=not converted (imported from True/False boolean)
total_ads	INT	Number of ads/PSAs shown to user (min:1, max:2065, mean:24.82, median:13)
most_ads_day	VARCHAR(10)	Day of week when user saw the most ads (Monday–Sunday)
most_ads_hour	TINYINT	Hour of day (0–23) when user saw the most ads

05 TOOLS & TECHNOLOGIES USED

Tool	Role	Key Actions
MySQL 8.0	PHASE 1 — Database & Analysis	CREATE TABLE, LOAD DATA INFILE, 7 queries, 1 VIEW
MySQL Workbench	MySQL GUI Client	Query execution, result verification, schema view
Power BI Desktop	PHASE 2 — Dashboard	MySQL connection, Power Query, 8 DAX measures, 2-page dashboard
DAX	Power BI Calculation Language	8 measures: conversion rates, lift %, group-filtered metrics
Sync Slicers	Power BI Feature	Test Group and Day slicers synced across Page 1 and Page 2
JSON Theme	Power BI Custom Theme	Digital Marketing Purple & Cyan — dark canvas, vibrant accents

Note: Excel was assessed and determined NOT required — zero data quality issues found in EDA (no nulls, no duplicates, no formatting problems, no date columns). CSV imported directly to MySQL.

06 PHASE 1 — MYSQL: IMPORT & ANALYTICAL QUERIES

Database & Table Creation

Created marketing_project database and marketing_ab table with 6 typed columns. user_id defined as INT PRIMARY KEY ensuring uniqueness. converted defined as TINYINT(1) storing 0/1 integers.

LOAD DATA LOCAL INFILE — Single Command Import

- Used @skip variable to silently discard the 'Unnamed: 0' index column (column 1 in CSV) without creating a corresponding table column.
- Used @conv variable + IF(@conv = 'True', 1, 0) to convert boolean text 'True'/'False' to TINYINT 1/0 — eliminating any need for post-import UPDATE statements.
- Imported 588,101 rows in 6.922 seconds with 0 warnings, 0 errors, 0 skipped rows — zero re-imports required.
- OPT_LOCAL_INFILE=1 was already configured in MySQL Workbench from Project 4 — no setup changes needed.

7 Analytical Queries + Results

Query	Purpose	Key Result
Q1 — Overall KPIs	Total users, conversions, overall conversion rate, avg/max ads	588,101 users · 14,843 conversions · 2.5239% rate · 24.82 avg ads · 2,065 max ads
Q2 — A/B Group Comparison	Conversion rate, user count, avg ads per test group	ad: 2.5547% (564,577 users) · psa: 1.7854% (23,524 users) · 43.09% lift

Query	Purpose	Key Result
Q3 — Conversion by Day	Conversion rate and volume for each day of week	Monday best: 3.2812% · Saturday worst: 2.1051% · Friday highest volume: 92,608
Q4 — Conversion by Hour	Hourly conversion rate across 24 hours (0–23)	Peak: Hour 16 at 3.0772% · Worst: Hour 2 at 0.7313% · Prime window: 14:00–17:00
Q5 — Ads Exposure Buckets	Conversion rate by total ads seen (6 buckets)	1-10 ads: 0.33% → 101-200 ads: 17.37% — 53x improvement with exposure
Q6 — Best Day+Hour Combos	Top 15 day+hour combinations by conversion rate (≥100 users)	Saturday 06:00: 4.91% · Tuesday 16:00: 4.53% · Monday 14:00: 4.44%
Q7 — CREATE VIEW	vw_ab_summary — grouped by group, day, hour, ads_bucket	Pre-aggregated 7-column view used directly in Power BI for performance

07 PHASE 2 — POWER BI: CONNECTION & DASHBOARD

8 DAX Measures Created

Measure	DAX Formula	Result
Total Users	COUNTROWS(marketing_ab)	588,101
Total Conversions	SUM(marketing_ab[converted])	14,843
Conversion Rate	DIVIDE([Total Conversions],[Total Users])*100	2.5239%
Ad Group Users	CALCULATE([Total Users], test_group="ad")	564,577 (96%)
PSA Group Users	CALCULATE([Total Users], test_group="psa")	23,524 (4%)
Ad Conversion Rate	CALCULATE([Conversion Rate], test_group="ad")	2.5547%
PSA Conversion Rate	CALCULATE([Conversion Rate], test_group="psa")	1.7854%
Conversion Lift %	DIVIDE([Ad CR]-[PSA CR],[PSA CR])*100	43.09%

2-Page Dashboard (16 Visuals)

Page	Visual Type	Key Insight
P1	Total Users KPI Card	588,101 total test participants
P1	Total Conversions KPI Card	14,843 total conversions
P1	Conversion Rate KPI Card	2.52% overall rate
P1	Conversion Lift % KPI Card	43.09% — ads outperform PSA
P1	Ad Conversion Rate KPI Card	2.55% (ad group)
P1	PSA Conversion Rate KPI Card	1.79% (control group)

Page	Visual Type		Key Insight
P1	Conversion Rate Comparison Clustered Bar		ad bar vs psa bar — visual gap
P1	Ad vs PSA Group Split Donut		564K (96%) ad · 23K (4%) psa
P1	Slicer: Test Group		Dropdown — filters all visuals
P1	Slicer: Day of Week		Dropdown — filters all visuals
P2	Conversion Rate by Day Clustered Bar		Monday 3.28% → Saturday 2.11%
P2	Conversion Rate by Hour Line Chart		Peak 14:00–17:00 window visible
P2	Conversion by Ads Bucket Clustered Bar		53x lift from low to high exposure
P2	Day vs Hour Heatmap Matrix		Best combos: Tue 16:00, Mon 14:00
P2	Slicer: Hour Range Between		0–23 range slider
P2	Slicer: Test Group		Synced from Page 1

08 KEY INSIGHTS & RESULTS

A. Core A/B Test Result — Campaign Effectiveness

- Ad campaign generated a statistically meaningful 43.09% conversion lift over PSA control — ad group converted at 2.5547% (14,423 conversions from 564,577 users) vs PSA group at 1.7854% (420 conversions from 23,524 users) — an absolute difference of 0.7693 percentage points confirming the ad campaign drives measurable incremental conversions.
- Both groups received virtually identical average ad exposure (ad: 24.82 avg ads · psa: 24.76 avg ads) — confirming the only difference between groups was ad content type, making the 43.09% conversion lift directly attributable to ad creative effectiveness rather than exposure frequency differences.

B. Optimal Ad Timing — Day of Week

- Monday delivered the highest conversion rate at 3.2812% (2,857 conversions from 87,073 users) — 55.9% above the lowest-performing day Saturday (2.1051%) — making Monday the highest-priority targeting day for maximising conversion efficiency.
- Tuesday 2.9840% ranked second; Wednesday 2.4942% third — establishing a clear weekday superiority pattern where Monday–Wednesday collectively outperform Thursday–Saturday by a consistent margin, suggesting workplace-hours intent is a primary driver of conversion.
- Friday had the highest user volume (92,608 users) but ranked 5th in conversion rate (2.2212%) — demonstrating that traffic volume and conversion efficiency are inversely related on this day, making Friday a poor choice for conversion-focused campaigns despite audience size.

C. Optimal Ad Timing — Hour of Day

- Hour 16 (4 PM) delivered the peak conversion rate at 3.0772% — establishing a 14:00–17:00 afternoon window as the prime ad delivery period; hours 20 (2.9803%), 21 (2.8923%), 15 (2.9653%), and 17 (2.8210%) rounded out the top 5.

- Hour 2 (2 AM) was worst at 0.7313% — 76.2% below the peak hour rate; hours 1 (1.2911%), 3 (1.0452%) and 7 (1.8111%) also significantly underperformed — defining a 01:00–07:00 blackout window where ad delivery generates minimal conversion return.
- Two distinct high-performance windows identified: Afternoon peak (14:00–17:00 averaging 2.90%) and Evening peak (19:00–22:00 averaging 2.82%) — campaigns should concentrate budget delivery in these 7 hours rather than spreading uniformly across all 24 hours.

D. Ad Exposure Frequency Effect

- Strong positive correlation confirmed between ad exposure and conversion — users seeing 1-10 ads converted at only 0.33% (858 conversions from 260,775 users) while users seeing 101-200 ads converted at 17.37% (2,972 from 17,112) — a 52.7x conversion rate improvement from minimum to optimal exposure.
- Exposure effectiveness peaks at 101-200 ads (17.37%) with slight decline at 200+ ads (15.57%) — suggesting diminishing returns beyond ~200 ad exposures; optimal frequency targeting window is 51-200 ads per user where conversion rates exceed 11% consistently.

E. Best Day + Hour Targeting Combinations

- Saturday 06:00 achieved the highest combination rate at 4.9057% (13 conversions from 265 users) — though low volume limits statistical confidence; Tuesday 16:00 emerged as the most reliable high-performance combination at 4.5284% across 4,262 users with 193 conversions — statistically robust.
- Monday dominated the top combinations — Monday 14:00 (4.4357%, 281 conversions, 6,335 users), Monday 15:00 (4.2490%, 284 conversions, 6,684 users), Monday 16:00 (3.7043%, 217 conversions, 5,858 users) — confirming Monday afternoon as the single most valuable ad delivery window.

F. Complete KPI Summary

KPI	Value	KPI	Value
Total Users	588,101	Total Conversions	14,843
Overall Conv Rate	2.5239%	Ad Conv Rate	2.5547%
PSA Conv Rate	1.7854%	Conversion Lift	43.09%
Ad Group Users	564,577 (96%)	PSA Group Users	23,524 (4%)
Ad Conversions	14,423	PSA Conversions	420
Avg Ads (Ad Group)	24.82	Avg Ads (PSA Group)	24.76
Best Day	Monday — 3.2812%	Worst Day	Saturday — 2.1051%
Best Hour	Hour 16 — 3.0772%	Worst Hour	Hour 2 — 0.7313%
Best Day+Hour	Tuesday 16:00 — 4.5284%	Best Low Vol Combo	Saturday 06:00 — 4.91%
Best Exposure	101-200 ads — 17.37%	Lowest Exposure	1-10 ads — 0.33%
Exposure Lift	52.7x (low → optimal)	Max Ads Shown	2,065

09 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Determining Whether Excel Was Necessary

- Problem: Previous projects (especially Project 4) required significant Excel pre-processing before MySQL import — causing multiple errors. The risk was assuming Excel was either always needed or always skippable without proper investigation.
- Solution: Conducted systematic EDA before writing a single instruction — checked nulls (0), duplicates (0), duplicate user IDs (0), data types, value ranges, and formatting issues. The analysis conclusively showed zero data quality issues requiring Excel intervention, saving the entire Excel phase and its associated potential errors.

Challenge 2 — 'Unnamed: 0' Index Column in CSV

- Problem: The CSV file contained a redundant sequential index column ('Unnamed: 0') as its first column — a common pandas export artifact. Including it would create a meaningless column in MySQL and waste storage; excluding it manually would require Excel preprocessing.
- Solution: Used `LOAD DATA LOCAL INFILE` with a `@skip` variable placeholder for the first column in the column list — (`@skip`, `user_id`, `test_group`, `@conv`, ...). The `@skip` variable receives the value but is never assigned to any table column, silently discarding it without any pre-processing.

Challenge 3 — Boolean True/False vs MySQL TINYINT

- Problem: The 'converted' column in the CSV contained 'True'/'False' text strings (Python boolean export format). MySQL TINYINT columns cannot accept these text values — importing would either fail with data type errors or store incorrect values.
- Solution: Used a second variable `@conv` in the `LOAD DATA` command column list, then `SET converted = IF(@conv = 'True', 1, 0)`. This converts 'True' → 1 and 'False' → 0 at the point of import — no post-import UPDATE required, no separate processing step needed, verified with `SELECT DISTINCT converted` returning only 0 and 1.

Challenge 4 — DAX CALCULATE for Group-Filtered Measures

- Problem: Creating separate DAX measures for Ad Conversion Rate and PSA Conversion Rate required filtering the base Conversion Rate measure to only one test group — a pattern not used in previous beginner projects that only used SUM/AVERAGE/COUNT.
- Solution: Used DAX `CALCULATE` function with a filter argument: `CALCULATE([Conversion Rate], marketing_ab[test_group] = "ad")`. `CALCULATE` overrides the current filter context with the specified filter — enabling the measure to compute conversion rate exclusively for the ad group regardless of any slicer selections, making it reliable across all dashboard interactions.

Challenge 5 — Ads Bucket Sorting Alphabetically in Visual

- Problem: The ads bucket chart (Visual 13) displayed buckets in alphabetical order (01-10, 101-200, 11-25, 200+...) rather than the logical numeric progression from low to high exposure — making the positive exposure-conversion trend difficult to read.
- Solution: Prefixed bucket names with sort keys in the SQL CASE statement ('01 — 1 to 10 ads', '02 — 11 to 25 ads'...) so alphabetical sort naturally produces the correct numerical order — a standard SQL-to-BI visual sorting technique that works without requiring Power BI column-by-column sort configuration.

10 SKILLS LEARNED & TAKEAWAYS

- EDA-First Methodology: Performing systematic EDA before any tool selection — determining Excel was unnecessary by checking nulls, duplicates, formatting, and data types — preventing hours of unnecessary preprocessing and potential import errors.
- Advanced LOAD DATA INFILE: Used @variable placeholders to skip columns and perform inline data transformations (boolean → integer) within the LOAD DATA command itself — eliminating post-import cleanup steps entirely.
- A/B Test Statistical Interpretation: Understood the distinction between absolute conversion difference (0.77pp) and relative lift (43.09%) — learned that the same result can be presented very differently and both metrics serve different stakeholder audiences.
- DAX CALCULATE Function: Implemented context-modifying CALCULATE() to create group-specific measures (Ad Conversion Rate, PSA Conversion Rate) that respond correctly to all slicer interactions — a fundamental DAX pattern used in virtually every real-world Power BI project.
- SQL Prefix-Based Sorting: Applied numbered prefix pattern ('01 —', '02 —') to CASE statement labels ensuring correct sort order in Power BI visuals without requiring manual sort configuration — a reusable technique for any categorical bucketing.
- Heatmap Design with Matrix Visual: Built a Day × Hour conversion rate heatmap using Power BI's Matrix visual with conditional background color formatting — creating a visual that immediately reveals high-performance targeting windows not visible in single-dimension charts.
- Exposure Frequency Analysis: Quantified the dose-response relationship between ad exposure and conversion — identifying the 52.7x lift from minimum to optimal exposure and the 101-200 ads sweet spot — translating data into actionable frequency capping recommendations.

11 CONCLUSION

- Conducted complete EDA confirming zero data quality issues, determined Excel was NOT required, and imported 588,101 marketing A/B test records directly into MySQL via a single LOAD DATA LOCAL INFILE command in 6.922 seconds — skipping the row-index column and converting boolean values to integers inline with zero errors.
- Executed 7 analytical SQL queries and created vw_ab_summary VIEW — quantifying a 43.09% ad campaign lift (2.5547% vs 1.7854%), identifying Monday as the best day (3.28%), Hour 16 as the best hour (3.08%), and Tuesday 16:00 as the highest-volume high-performance targeting combination (4.53% across 4,262 users).
- Discovered a 52.7x conversion rate improvement from minimal exposure (1-10 ads: 0.33%) to optimal exposure (101-200 ads: 17.37%) — a critical frequency targeting insight demonstrating that ad exposure frequency is the strongest single predictor of conversion in this dataset.
- Built 8 DAX measures and a 2-page Power BI dashboard with 16 visuals including a Day × Hour conversion heatmap, ads exposure bucket chart, group comparison visuals, and Sync Slicers — all connected live to MySQL — delivering campaign optimization recommendations across timing, exposure, and audience segmentation dimensions.
- First project in the portfolio to completely skip Excel based on EDA-driven decision-making — demonstrating the analytical judgement to select only the tools genuinely required by the data, rather than following a fixed template. Project 5 of 9 — second Intermediate project completed.